ripwire

The ripgrep of AI context.

A zero-runtime-dependency C++23 CLI that maps any codebase into a ranked, deterministic call graph for coding agents — and puts a tripwire on every claim it emits.

rip — the speed half

Structural answers in tens of milliseconds, from a call graph it built itself.

wire — the honesty half

Unprovable totals ship labelled as floors. A zero means none found — never none exists.

Apache-2.0 · single binary · hermetic build (proven with the network off) · 15 vendored tree-sitter grammars

// the problem

Your agent reads the whole library to answer one question

Blind grep, then whole-file reads. Most of what enters the context window is never used — it is paid for anyway, on every step of every task.

And bigger context is not better context: irrelevant text actively degrades the answer. The job is selection — a small, ranked, structural map beats a pile of open files.

0.114 s

median answer, warm index — latest head-to-head round, N = 60

−39.4%

token ceiling vs the un-routed baseline — LocBench cost ledger, N = 243

Speed caveat travels with the number: ripwire answers from a warm, pre-built index; competitor medians include their per-question work. Same instances, same gold, same metric code.

bench/headtohead/r2-2026-08-03/ · bench/locbench/ — every figure on this deck names its instrument

// one binary, the whole pipeline

Crawl → parse → resolve → rank → emit. Deterministic, end to end.

crawl	→	parse	→	resolve	→	rank	→	emit
the tree, no VCS needed		tree-sitter, 15 vendored grammars		references → call graph		Personalized PageRank		minified XML, one line

byte-identical	two runs, same bytes — a gate on every push, not a tendency; warm equals cold
zero runtime deps	CMake + a C++23 compiler; builds with the network off — vendored everything
11 languages	C/C++, ObjC, Python, TS/JS, Java, Go, Rust, Ruby, Swift, C#, Bash — plus JSON and Metal (a C++ dialect)
agent-native	an MCP server and 124 long flags behind one `--help` that is always the authority

// every feature, one screen

Seven families of questions it answers

understand cold	*What is this repo, and what matters in it?*	`--for` `--tree` `--lego` `--exemplar` `--recall` `--top-k` `--token-budget`
navigate	*Who calls this? Safe to change? Which tests?*	`--callers` `--callees` `--uses` `--impact` `--path` `--connect` `--affected` `--situ` `--test-gate` `--grep`
detail ladder	*Show me more — but only where it pays.*	`--detail` `--pack-signatures` `--outline` `--expand` `--compress`
quality & risk	*Where is the risk, and did I just add some?*	`--hotspots` `--clones` `--metrics` `--deps` `--lint` `--quality-delta` `--edit-check` `--pr-context` `--merge-scout`
self-diagnosis	*Is my setup actually working?*	`--doctor`
security	*Is this agent skill file safe to install?*	`--scan-skill` `--scan-skills`
knobs & modes	*Shape, format, cache, budget.*	`--json` `--format` `--mcp`

`ripwire --help` is generated from the binary's own flag table — 124 long flags; docs/COMMANDS.md documents 101 of them with real recorded output

// reach for it at the moment, not the manual

The reflexes: which verb fires when

Landing cold on a task

`--for="<task>"` · `--pack-task`

ranked, budgeted context in one call

Stack trace in hand

`--from-trace`

frames mapped to symbols, innermost body included

“Is it safe to change X?”

`--impact` · `--uses`

the whole blast radius, not just 1-hop callers

Just edited a symbol

`--edit-check`

did the contract change, who breaks — ~ms, warm

About to write a helper

`--exemplar` · `--grep`

imitate the house pattern; catch the duplicate early

Landing several branches

`--merge-scout`

pairwise conflict sites + a landing order

Before you call it done

`--quality-delta` · `--test-gate`

only what you made worse; exactly which tests to run

Reviewing a diff

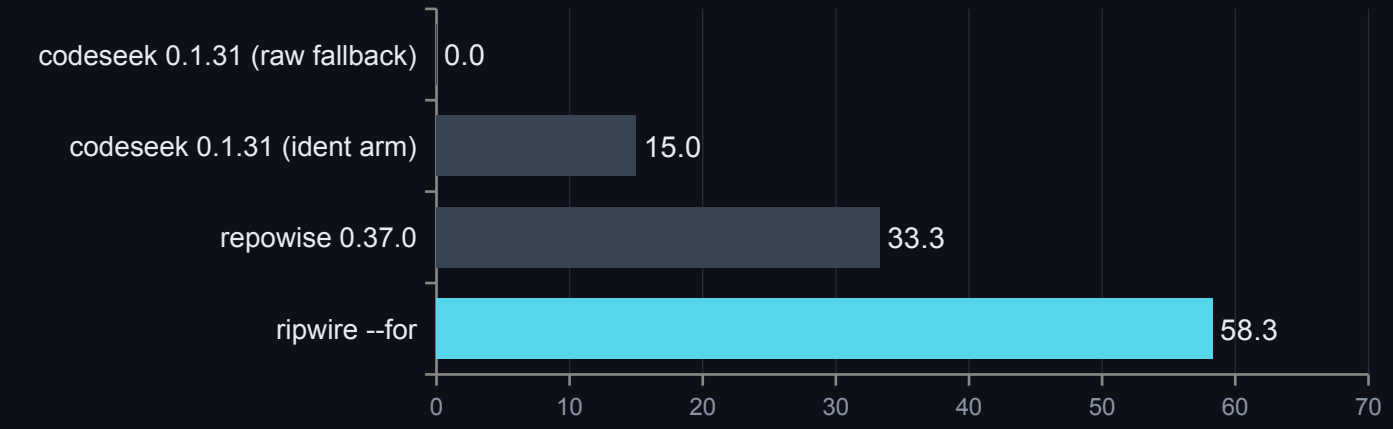
`--pr-context`

per-file blast radius, tests-to-run, hotspot flags

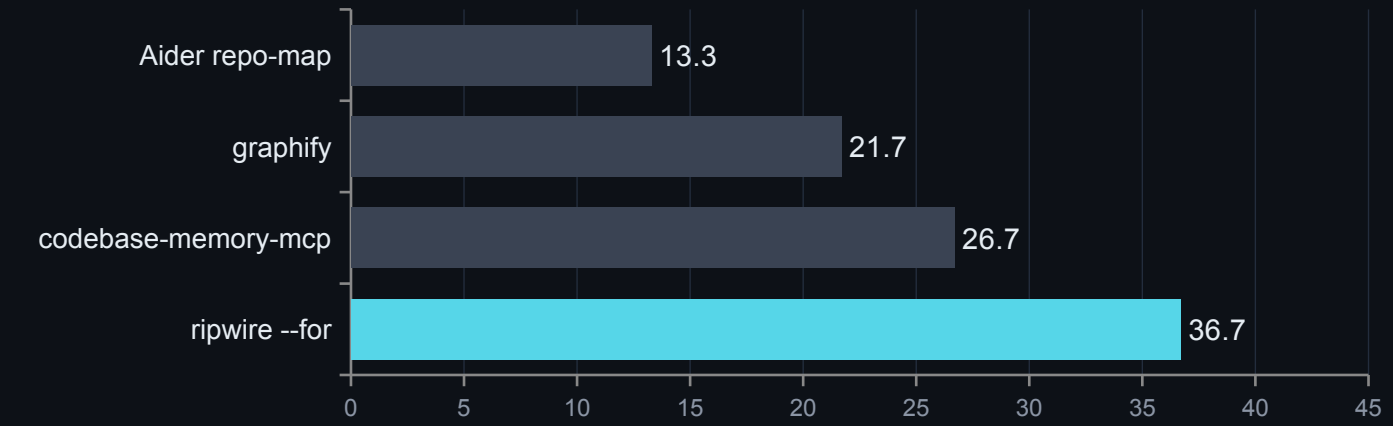
// same instances, same gold, same metric code

Head-to-head: every round, every tool

Round 2 (2026-08-03) — strict file@10, N = 60



Round 1 — strict file@10, N = 60 (own binary + evaluator; rounds are not number-comparable)



58.3%

strict file@10 — every gold file in the top ten
N = 60 paired, zero exclusions · 17–2 paired vs repowise

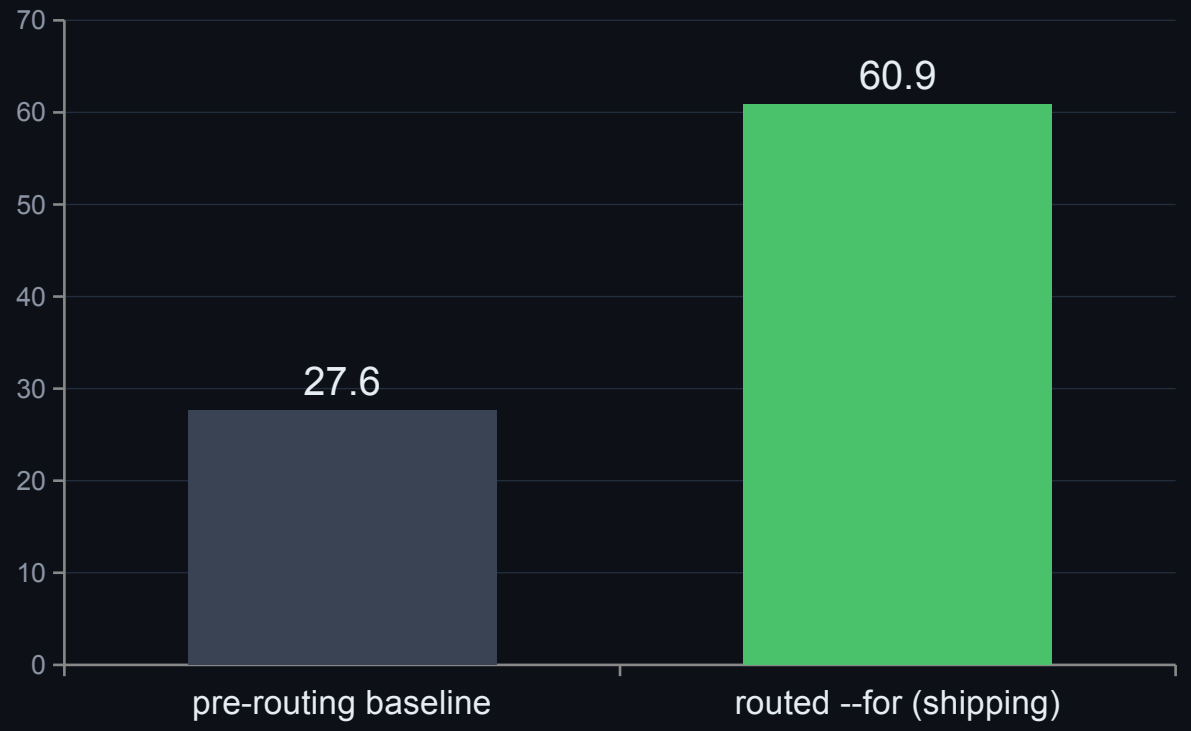
Median wall (query, warm): ripwire 0.114 s · repowise 1.14 s (incl. a per-query server spawn) — round 1: Aider 2.5 s · graphify 5.8 s, cold per run.
codeseek's raw arm: 0 results on 60/60 — a query-protocol boundary, not its embedder mode. Vexp and CodeIndexer were excluded by free-tier caps, not beaten.

Round 3, different instrument: headroom (the compression layer) passed code through byte-identical; ripwire answered at 7.3% of the naive baseline's tokens.

bench/headtohead/REPORT.md (r1) · r2-2026-08-03/ · r3-headroom-2026-08-03/ · docs/EVALS.md §2 — an adversarial VERIFIER.md ships with each round

// held-out, repo-disjoint, frozen dataset

Routing the ranker: +33 points, measured properly



+33.33pp — paired delta, 243 held-out instances in 78 repository clusters

+25.00pp — clustered-bootstrap 95% lower bound — the honest floor

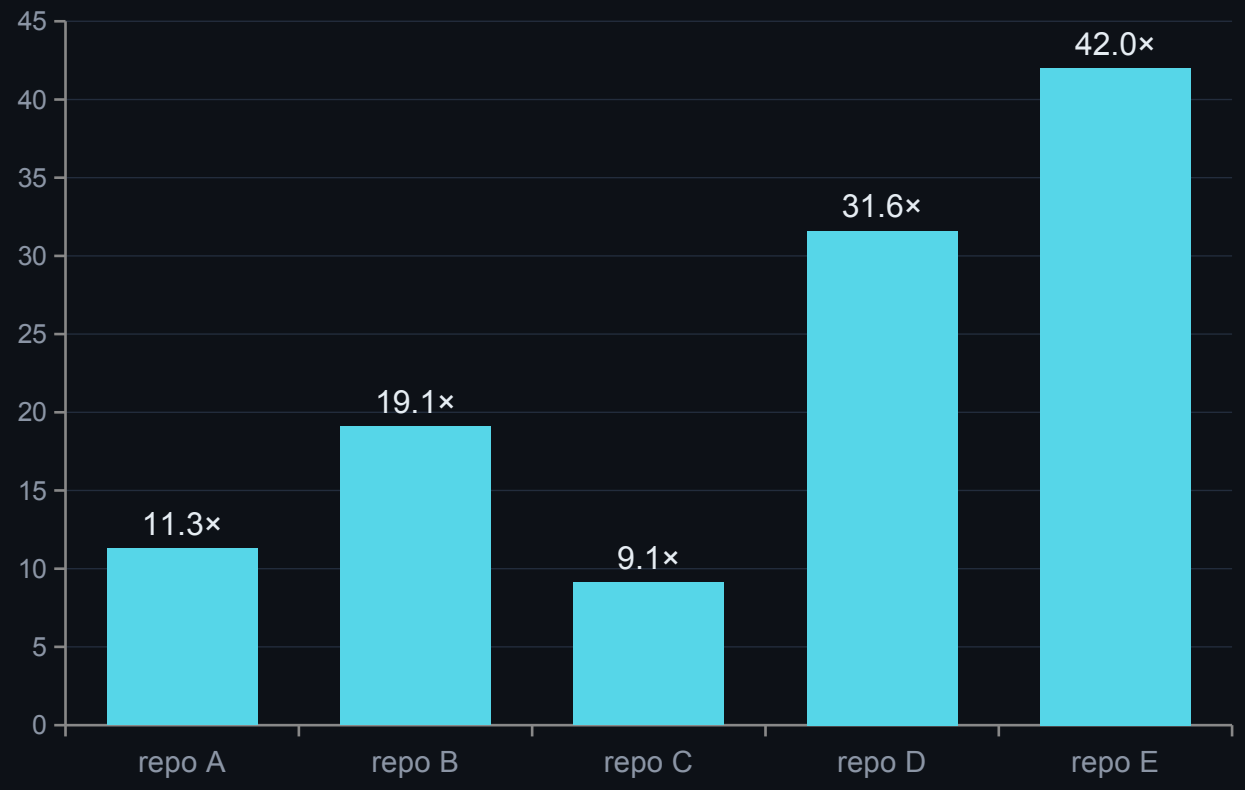
−39.4% — token ceiling for that gain

+3.4% — warm latency for that gain

bench/locbench/ — czlll/Loc-Bench_V1, 560 rows, JSON SHA-256 pinned in dataset.lock; metric code = LocAgent's strict Acc@k

// compiled, not interpreted

How much faster: same algorithm, 9-42× the speed



Five repositories, tag caches cleared each run, minimum of N runs. Same pipeline on both sides — tree-sitter parse then PageRank — so the gap is compiled versus interpreted.

1,560-file repository

~1 s cold · 0.18 s warm

Aider's map of the same tree: 40 s cold

Where the second goes (self-profiled)

parse ~780 ms → 33 ms warm (24×, the cache)
resolve + PageRank together: ~21 ms

The ranking is the cheap part — the graph math was never the bottleneck.

bench/BENCHMARK.md (caveat: measured 2026-06-20 on a large private C++ corpus — historical, not publicly reproducible) · bench/PROFILE.md

// the performance sprint

Half the wait: work removed before work optimized

100 ms	20 ms	< 3 ms	green
cold C++ run — was 190 ms, −47%	warm cache run — was 40 ms, half	graph + rank + emit — below the noise floor	determinism, ASan, det-gate and cache gates — unchanged

file ingest	FILE* whole-file reads; CSV counting became a byte scan — iostream out of the hot path
--no-cache	skips contentHash64() entirely when no lookup or save can use it — pure waste removed
capture gating	value-use capture runs only for --uses, --metrics, --for and --exemplar — side captures 404 → 57 ms
parallel parse	query compile overlaps parse scheduling; files schedule by size — ids stay deterministic
backlog cap	parsed-file backlog bounded at 4 files / 8 MiB per worker — memory pressure stays sane

pre-release performance sprint — measured on the private ~1,500-file C++ corpus (historical, not publicly reproducible); same map contract: byte-identical output before and after

// where the time goes now

Faster at scale — and the long pole is the parser

radix edges

numeric keys for graph edges — ~6× edge sort at scale, on the production graph path

radix scores

score/id ordering made numeric — 5.8–8.5× in benchmarks, deterministic tie-break kept

timsort scratch

caller-owned scratch buffer, no per-frame allocation — best on nearly-sorted updates

pdqsort bench

added as the adversarial baseline; std::sort held up well — the baseline keeps us honest

S+tree hot map

bounded-scratch dynamic map: 20.4 ms vs 68.3 ms lookup — a fixed-cap guard reports overflow

hash lookup

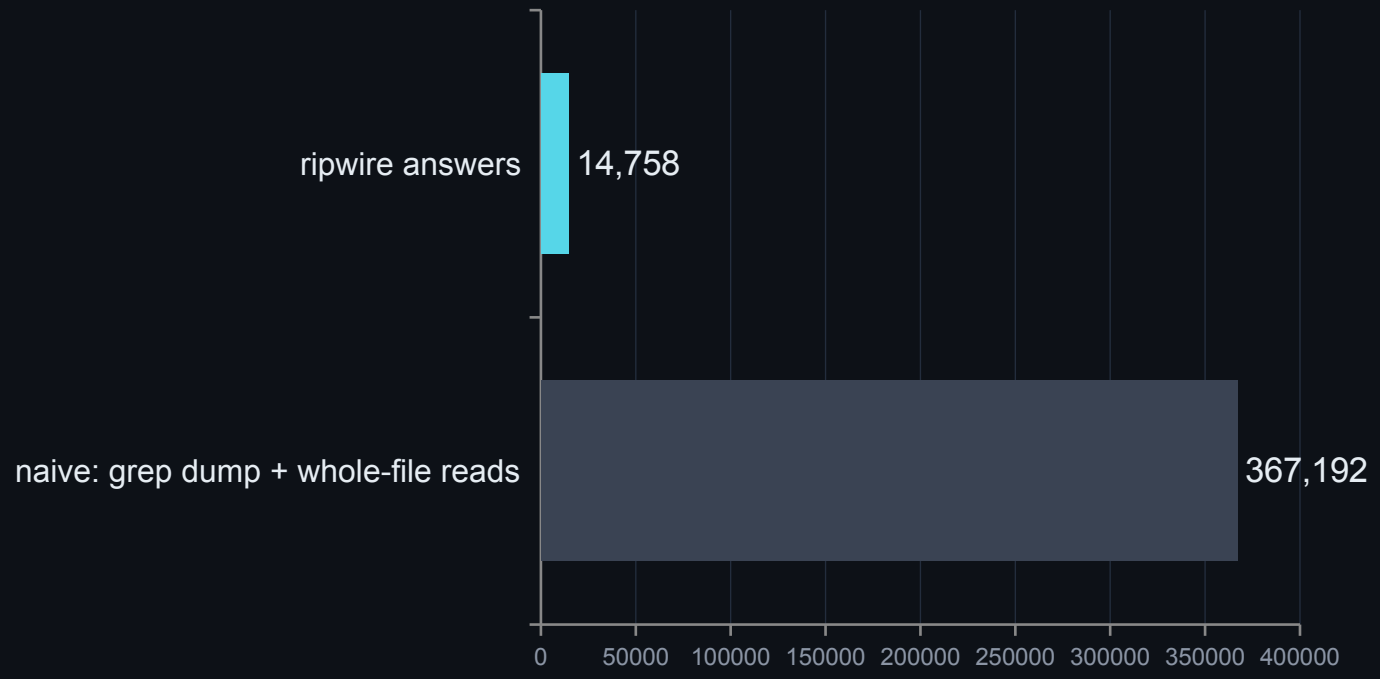
unordered_dense HashMap, reserve() before ingest — no std::unordered_map on any hot path

Cold profiles are now tree-sitter parse + tag-query capture, not graph math. Demand-driven captures already cut 404 → 57 ms; the next wins are query- and grammar-specific. *Same stuff it needs to do — just a lot less waiting.*

pre-release performance sprint — sort/index wins matter most past cache scale; determinism and sanitizer gates stayed green throughout

// the token bill, cut

Six real agent questions: 24.9× fewer tokens than the naive read



96.0%

fewer tokens — 14,758 vs 367,192, tiktoken cl100k_base, six questions: who-calls, orientation, blast radius …

The anti-headline, published next to it: **`--grep costs +19.7% tokens (−11.2%)`** — not every verb is a reducer, and the docs say which.

Caveat carried with every figure on this slide: measured 2026-06-20 on a large private C++ corpus — historical, private, not publicly reproducible. The public, gated companions are the 67.0% element-byte reduction and LocBench's −39.4% ceiling.

bench/BENCHMARK.md · docs/EVALS.md §5 — the baseline is what an agent does WITHOUT the tool

// the cost lever

Smaller answers that are also better answers

67.0%

fewer element bytes at top-50 — the --pack-signatures ladder, root-neutralised measurement

Ask for a symbol by NAME and the router uses the name-exact ranker:

```
MRR  0.859 → 0.993
recall@1  76.7% → 98.7%
name-shaped queries on src/, held-out labels
```

This figure survived its own audit: three corrections deep, re-derived root-neutrally after the original was shown to depend on how the corpus path was spelled. A gate re-derives it against every fresh capture.

And the pollution metric — fixture and generated paths contaminating results — is driven to 0.0% on the ranking lane while recall went UP, not down.

docs/EVALS.md §4–§5 — showcasecapturecheck re-derives the byte-reduction triple; recallevalcheck pins the ranking bars

// the half no other tool ships

The tripwire: honesty as an output contract

counts_floor="1"

call edges come from source text — dynamic dispatch adds no edge, so every count that cannot claim completeness is labelled a floor, in the output itself

a zero is a measurement

zero means none found, never none exists — and absent ≠ 0

truncation is disclosed

every cap, page seam and budget cut is named in the header before you read a row

refusals teach

a refused query names the flag, the problem, and a working example — never a bare error

```
$ ripwire . --callers=rankGraphTeleport

<callers of="rankGraphTeleport"
  defs="1" count="6"
counts_floor="1">
<s t="fn" n="runEval" .../>
<s t="fn" n="rankGraph" .../>
…
</callers>
```

docs/EVALS.md §8 — the list of numbers this project refuses to publish, each with the reason. The counterexample section is not an afterthought.

// how it stays true

Proven, not promised

334 gate scripts
the suite runs on every push — plus determinism, cache-transparency and golden contracts; the gate count itself is gated against the runner's own loop

byte-identical, always
two runs over the same tree produce the same bytes; warm equals cold. Enforced in CI, twice — Release AND a plain flavour, because NDEBUG once blinded a whole class of checks

differential refactoring
a refactor must prove it changed nothing observable: two binaries, hundreds of argv vectors, stdout + stderr + exit codes byte-identical

held-out labels, authored blind
eval labels were written by reading source before the ranker ever ran on them — so the eval is allowed to say the ranker is wrong. It has.

sanitizer wall
ASan + UBSan + integer + float-cast, no-recover; TSan separately; leak suppression is one pinned file

its own output is audited
the showcase capture is regenerated and adversarially re-read as a claims corpus — the methodology ships in docs/METHODOLOGY.md

// own the losses — they are in the README too

Where it loses, published with the wins

--grep costs more than it saves	+19.7% tokens / −11.2% — measured, in the docs, next to the flag it indicts
PageRank is the wrong co-change ranker	3.8% recall@5 against 40.3% for plain lexical — relatedness is lexical; importance is structural
a signature can outweigh its body	one symbol's signature + doc comment: 303 bytes; its full body: 158 — the compact form lost
strict multi-file localization stays hard	single-file gold 73.4% vs multi-file 18.2% (held-out); the same cliff on every corpus — open headroom, said plainly

A tool that publishes its counterexamples is a tool whose wins you can believe.

// built for the agent's seat

Wire it into Codex in one command

```
$ codex mcp add ripwire -- ripwire --mcp
```

plugin bundle also ships all 17 skills

30 MCP verbs

15 read verbs mirroring the CLI, 12 flagship reflexes (impact, uses, edit_check, from_trace, connect …), 3 span-addressed edit verbs with a safety contract

lazy-body handles

read verbs return signatures and a stable handle; the agent fetches a body only when it decides it needs one — names by default, bytes on request

17 agent skills

moment-matched workflows (orient, navigate, change-check, quality-bar …), bundled in the Codex plugin or installed by skills/install.sh

prompts/ — self-improvement

ten copy-paste orchestrator loops: run the same audit, eval and head-to-head machinery that built the tool, on your own repo

the MCP server exposes the same deterministic engine — one index, shared with the CLI, staleness-checked

// standing on giants

The research inside — classic and current

27 repositories + 27 papers folded · 220 more surveyed and labeled — every row with the lesson taken and where it lives: docs/LINEAGE.md

1976	**Cyclomatic complexity — McCabe**	the metric behind --hotspots
1994	**Okapi BM25 — Robertson · Spärck Jones**	the lexical ranker (twice: subtoken+body, whole-name)
1998	**Personalized PageRank — Page · Brin**	the headline importance signal
1999	**HITS hubs & authorities — Kleinberg**	the alternate lens (--rank-by=authority|hub)
2008	**Louvain modularity — Blondel et al.**	the module / community map
2009	**Reciprocal Rank Fusion — Cormack et al.**	deterministic signal fusion (--rank-by=rrf)

TDAD · arXiv 2603.17973

a static test-map cut agent-caused regressions 6.08% → 1.82%; prose TDD instructions alone made agents WORSE → the shape of --test-gate

What-to-Retrieve · arXiv 2503.20589

retrieval selection beats retrieval volume for coding agents → the selection-over-dumping thesis

LocAgent / Loc-Bench (2025)

the localization metric (strict Acc@k) and the frozen 560-instance dataset every accuracy number here is scored on

tree-sitter

the incremental GLR parsing substrate — 15 grammars vendored, one parser per thread

counts gated in-repo (readmedriftcheck arm E derives them from LINEAGE.md's own tables) — the lineage is the design rationale, not decoration

// sixty seconds to the first ranked map

Quickstart

```
git clone <repo> && cd ripwire
cmake -S . -B build && cmake --build build -j      # hermetic: works with the network off
./build/ripwire --help
```

`ripwire .`	the ranked map — start here on any unfamiliar repo
`ripwire . --for="cache invalidation"`	the task lens: what to touch, ranked
`ripwire . --callers=someFunction`	who calls it — with the honesty marker attached
`ripwire . --test-gate`	before you commit: exactly which tests must run

ripgrep for the speed. **tripwire** for the honesty. Apache-2.0.